Recap: Lecture 22

In Lecture 22, we

- Used separation of variable to find the heat distribution along a rod to be

$$u(x,t) = -\frac{4L}{\pi^2}\sum_{m=1}^{\infty}\frac{(-1)^m}{(2m-1)^2}\mathrm{e}^{-\alpha\omega_{2m-1}^2 t}\sin(\omega_{2m-1}x) \quad \text{where} \quad \omega_n = \frac{n\pi}{L}$$

- Saw that it smooths an initial kink
- This is a *formal solution*. We didn't discuss whether it is a genuine solution

Lecture 23

In Lecture 23, we will:

1. Apply separation of variables to another problem involving the 1D heat equation
2. Start a third problem that involves inhomogeneous BCs

Heat equation in 1D: Problem 2

Consider Problem 1 but where the rod is entirely insulated, so that heat cannot escape

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Governing PDE: 1D heat equation for temperature distribution $u(x,t)$ is

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad \text{for} \quad 0 < x < L$$

Initial condition:

$$u(x,0) = f(x) \quad \text{for} \quad 0 < x < L$$

Boundary conditions: Neumann BCs

$$u_x(0,t) = u_x(L,t) = 0 \quad \text{for} \quad t > 0$$

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This is identical to Problem 1:

$$u(x,t) = X(x)\,T(t)$$

$$\Rightarrow \quad X\,T' = \alpha\,X''\,T$$

$$\Rightarrow \quad \frac{T'}{\alpha\,T} = \frac{X''}{X} = \lambda$$

$$\Rightarrow \quad T' - \alpha\lambda\,T = 0 \quad \text{and} \quad X'' - \lambda\,X = 0$$

2. Apply the boundary conditions to find the eigenvalues and eigenfunctions. *This involves a systematic analysis of three possible cases for the constant of separation.*

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$\lambda = k^2 > 0$:

$$X'' - k^2\, X = 0 \quad \Rightarrow \quad X(x) = A\, \mathrm{e}^{k\,x} + B\, \mathrm{e}^{-k\,x}$$

$\lambda = 0$:

$$X'' = 0 \quad \Rightarrow \quad X(x) = A\,x + B$$

$\lambda = -k^2 < 0$:

$$X'' + k^2\, X = 0 \quad \Rightarrow \quad X(x) = A\, \cos(k\, x) + B\, \sin(k\, x)$$

3. Solve the remaining ODEs for each eigenvalue

$$\boxed{T' - \lambda T = 0}$$

$\lambda = 0$

$\lambda = -\omega_n^2$

4. Apply Superposition Principle to get a series general solution. *The series will have unknown coefficients to be determined.*

5. Apply the initial conditions to find the series coefficients.

$$u(x,0) = f(x) \quad \Rightarrow$$

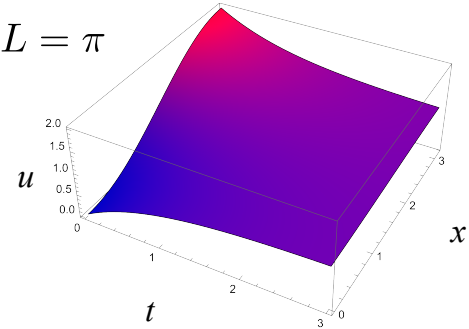
Let the IC be

$$u(x,0) = f(x) = 1 - \cos\left(\frac{\pi x}{L}\right)$$

The final solution is

$$u(x,t) = 1 - \mathrm{e}^{-\alpha\,\omega_1^2\,t}\cos(\omega_1\,x)$$

$$\alpha = 1, \quad L = \pi$$



Heat equation in 1D: Problem 3

Consider a rod which is initially at zero temperature. The temperature at one end is heated instantaneously and held at a higher constant temperature equal to the rod length, while the other is held at the initial temperature.

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$$u(x,t) = u_S(x) + u_T(x,t)$$
- u_S is the *steady-state solution*, which is independent of time
- u_T is the *transient solution*, such that, as $t \to \infty$,
$$u_T(x,t) \to 0 \quad \Rightarrow \quad u(x,t) \to u_S(x)$$

Steady state solution, $u_S(x)$

Obtain governing ODE

Substitute $u_S(x)$ into PDE to get

$$\frac{\partial}{\partial t}u_S(x) = \frac{\partial^2}{\partial x^2}u_S(x) \qquad \Rightarrow \qquad u_S'' = 0$$

Impose inhomogeneous BCs on $u_S(x)$

$$u_S(0) = 0 \qquad \text{and} \qquad u_S(L) = L$$

Transient solution $u_T(x,t)$: BCs and ICs

BCs are homogeneous for $u_T(x,t)$

$$u(0,t) = u_T(0,t) + u_S(0) = 0 \qquad \Rightarrow \qquad u_T(0,t) = 0$$

$$u(L,t) = u_T(L,t) + u_S(L) = L \qquad \Rightarrow \qquad u_T(L,t) = 0$$

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IC is a function of both $u(x,0)$ and $u_S(x)$

$$u(x,0) = f(x) = u_S(x) + u_T(x,0) \qquad \Rightarrow \qquad u_T(x,0) = g(x)$$

$$\text{where} \quad g(x) = f(x) - u_S(x)$$

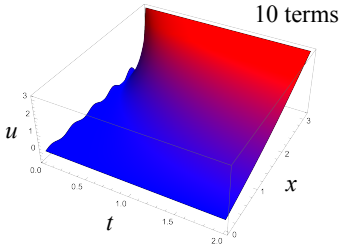
Problem 3: Solution

The full solution is

$$u(x,t) = u_S(x) + u_T(x,t) = x + \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\mathrm{e}^{-\alpha\,\omega_n^2\,t}\sin\left(\omega_n\,x\right)$$

$$\alpha = 1, \quad L = \pi$$

10 terms



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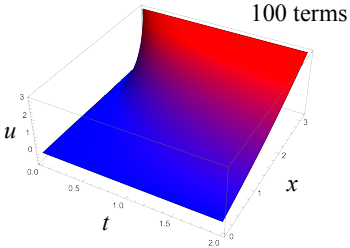
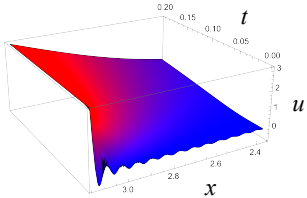
Zoom in on initial condition 100 terms